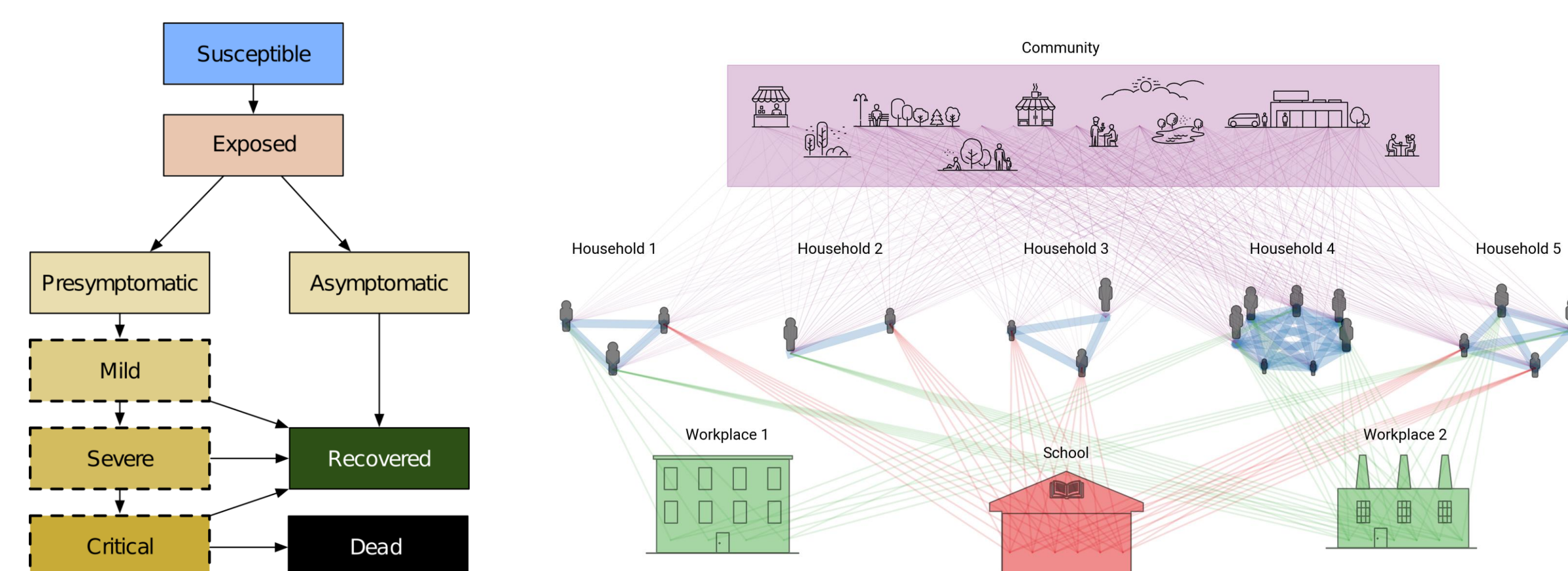


Introduction

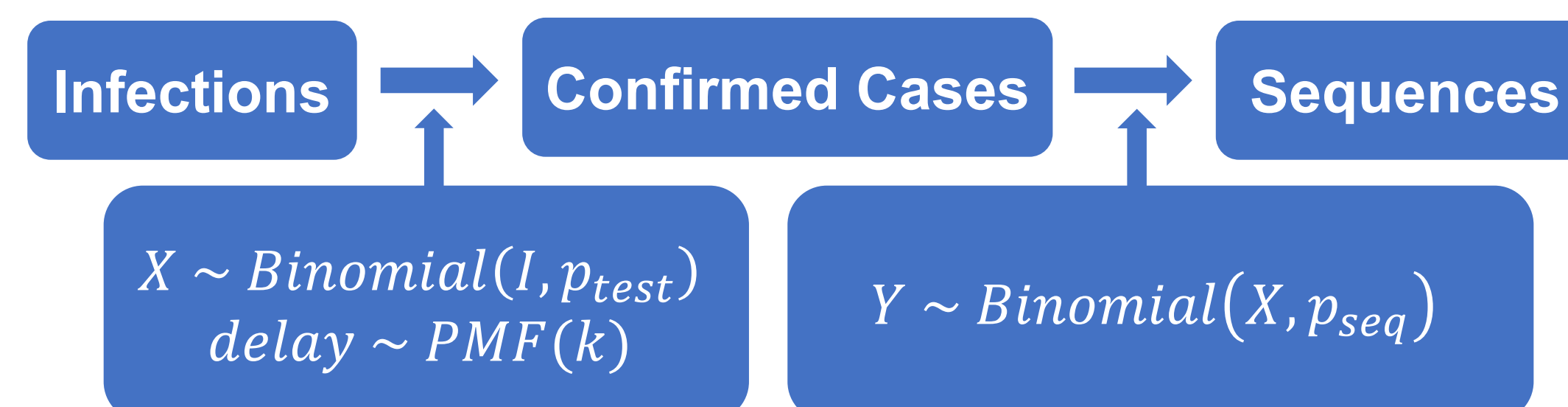
- Outbreak dynamics are only partially observable. Reported cases and sequenced samples represent only a small fraction of true infections.
- When outbreak dynamics are only partially observable, how well can different surveillance strategies recover the true epidemic, and how does this affect our ability to respond?**
- We simulate the full infection process and apply a sequencing-aware observation model under different surveillance strategies. This allows us to evaluate how various testing and sequencing policies capture true transmission dynamics and how different approaches influence public health interventions and outbreak control.

Methods

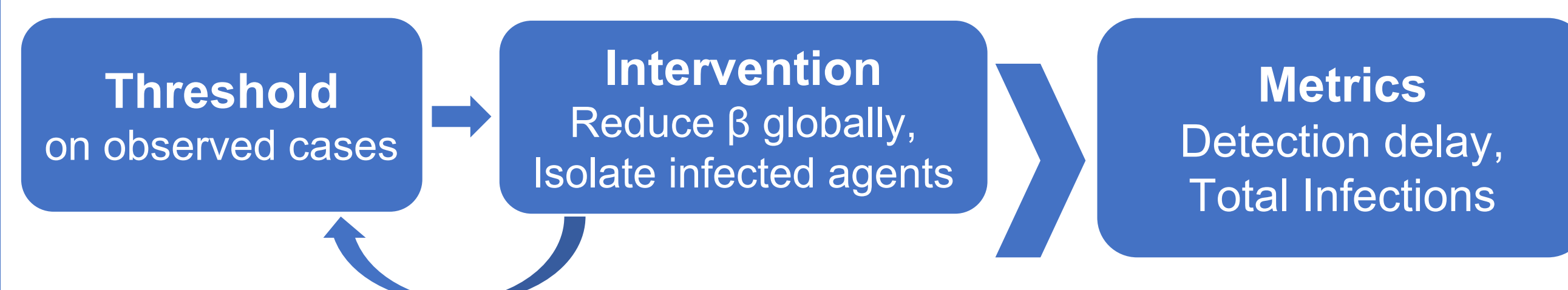
- What is the true epidemic we compare against?**
Agent-based simulator Covasim generating full ground-truth transmission: individual disease-progression states and who-infects-whom dynamics across structured contact networks.



- How do true infections become observable?**
We model how true infections become observed cases using an observation pipeline.

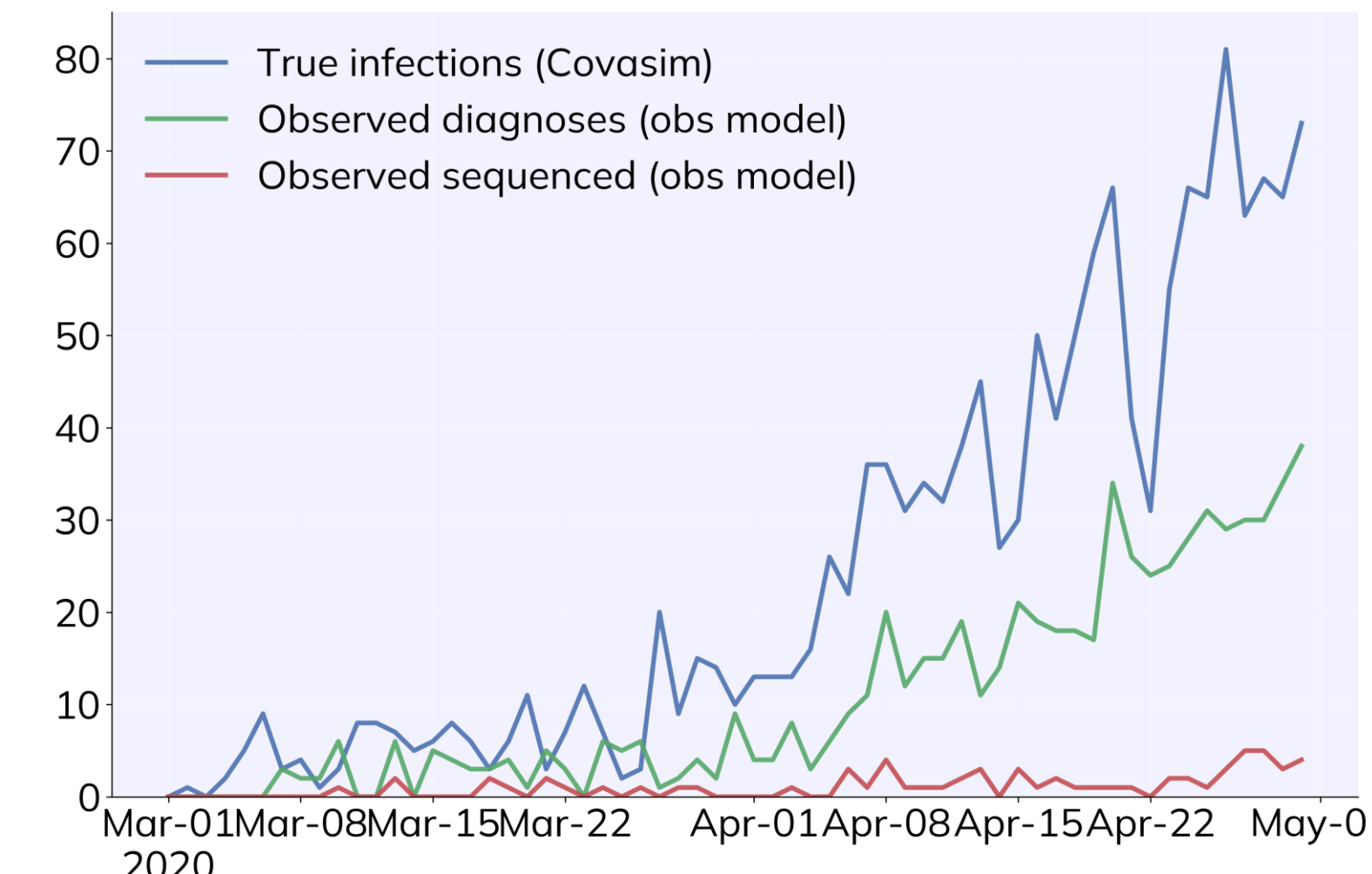


- How do surveillance strategies affect outbreak detection and control?**
We assess how testing and sequencing strategies shape detection.



Results

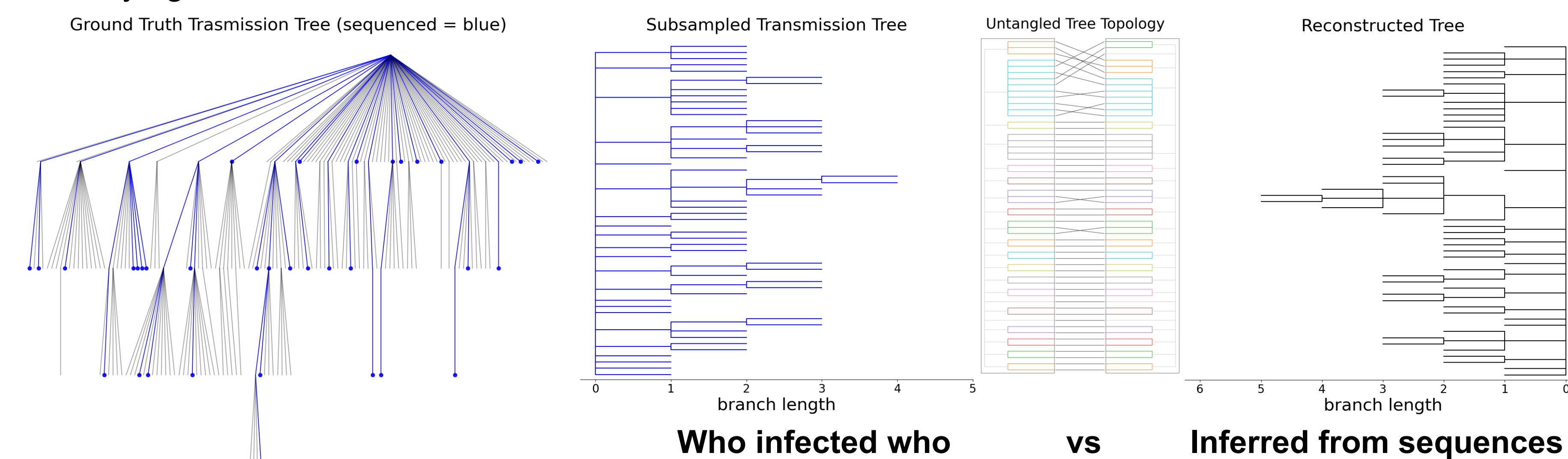
Reconstructing the Observable Epidemic



- Our model reconstructs the observable epidemic by identifying **which infections undergo diagnostic testing**, and **which diagnosed cases are selected for sequencing**, under each surveillance strategy.
- By recovering these two pathways, the model provides a realistic mapping from true infections to observed data, clarifying how surveillance design determines what becomes visible to public-health decision-makers.

Recovering Transmission Relationships from Sequenced Case Trees

We assessed how well phylogenetic trees reconstructed from sequenced cases recover the true underlying transmission tree.



Revealing Intervention Trade-offs with Our Observation Model

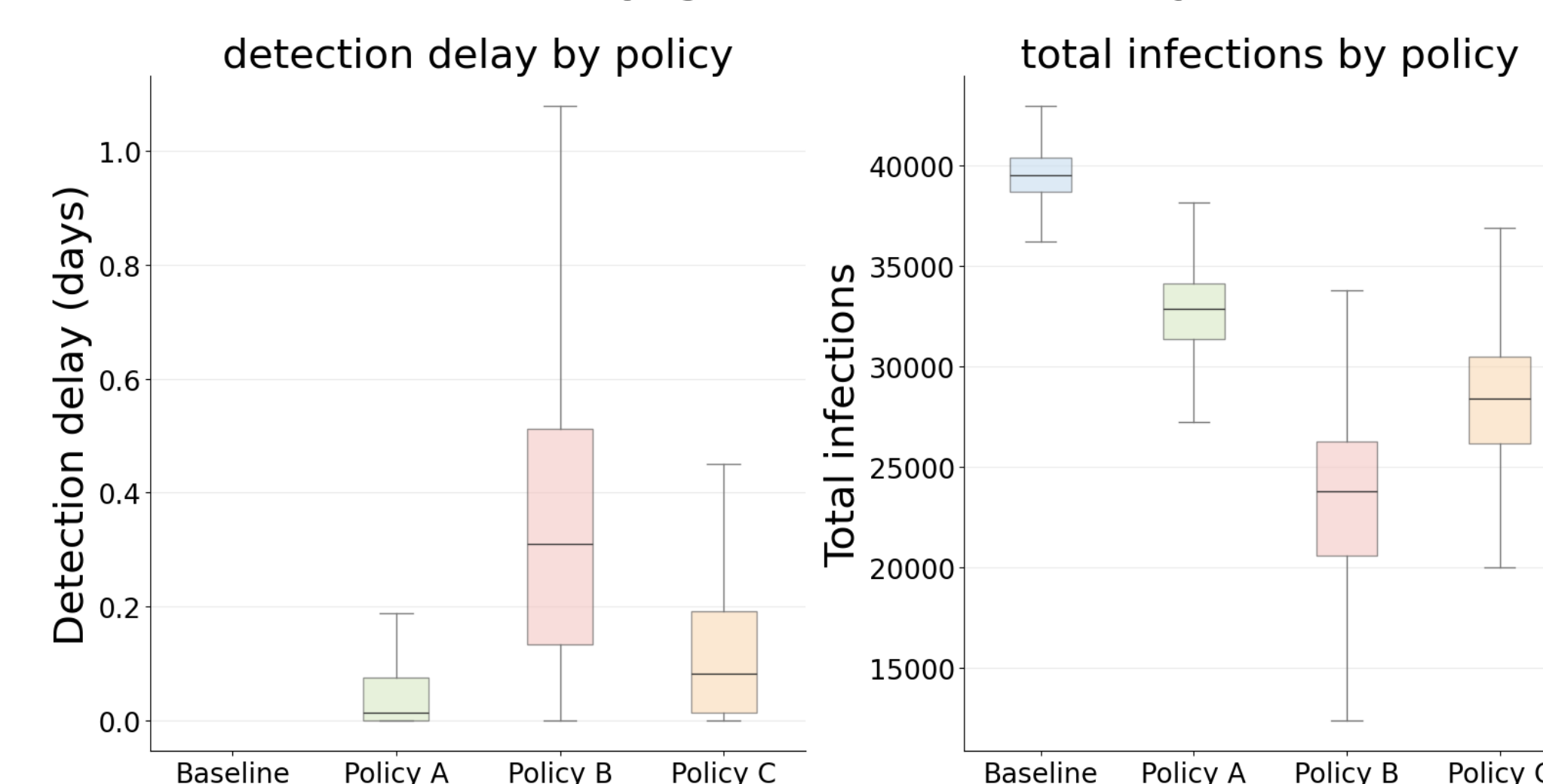
Intervention Strategies

Testing three policies:

- A (high testing, weak distancing)** - Costly high testing burden.
 - B (low testing, strong distancing)** - High social and economic cost from distancing.
 - C (balanced testing & distancing)** - More feasible than extreme strategies.
- Against a **no-intervention baseline**, running **500 simulations per scenario**.

Intervention Trigger

Policies activate when the **observed weekly growth rate** of diagnosed cases exceeds **1.1**.



Outcome: Policy B > Policy C > Policy A

Conclusion

- Our observation model simulates realistic surveillance by generating diagnosed cases and sequenced genomes through binomial sampling from the ground-truth epidemic. This allows us to reconstruct the observable epidemic trajectory that public-health systems would see under different surveillance designs.
- By subsampling genomes from the ground-truth transmission tree, we can examine how much sequencing is required to recover meaningful transmission relationships. This provides a practical way to evaluate the informativeness of genomic surveillance when only a fraction of infections can be sequenced.
- Together, these components reveal how surveillance choices shape the epidemic we are able to observe, how much transmission structure can be recovered from partial sequencing, and how these observable dynamics influence our ability to interpret and evaluate intervention strategies.

References

Kerr, Cliff C., Robert M. Stuart, Dina Mistry, Ruwan G. Abeyesuriya, Katherine Rosenfeld, et al. 2021. *Covasim: An agent-based model of COVID-19 dynamics and interventions*. *PLOS Computational Biology* 17 (7): e1009149. <https://doi.org/10.1371/journal.pcbi.1009149>

Schlegel, Philipp. *tanglegram: Enhanced dendrograms and tanglegrams*. GitHub repository. <https://github.com/schlegelp/tanglegram>.

Wong, Thomas K. F., Nhan Ly-Trong, Huaiyan Ren, Piyumal Demotte, Hector Baños, Andrew J. Roger, Edward Susko, Chris Bielow, Nicola De Maio, Nick Goldman, Matthew W. Hahn, Mario dos Reis, Le Sy Vinh, Gavin Huttley, Robert Lanfear, and Bui Quang Minh. 2026. *IQ-TREE 3: Phylogenomic inference software using complex evolutionary models*. *Molecular Biology and Evolution* 43 (5): msag117. <https://doi.org/10.1093/molbev/msag117>.

Acknowledgement

Supported by mentors Joshua I. Levy and Maryam Ahmadi Jeshvaghane, with additional contributions from Karthik Gangavarapu and Dylan Pilz, and further support from the Andersen Lab, the SRTI Internship Program, Theresa Hill, and the Bill & Melinda Gates Foundation.